

Optimizing Inventory Management and Sales Performance in a Retail Pharmacy using Data-Driven Techniques

A Final report for the BDM capstone Project

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Contents

1	Executive Summary and Title	2-3
2	Detailed Explanations of Analysis Process/ Methods	3-10
3	Results and Findings	10-20
4	Interpretation of Results and Recommendations	21-24

Executive Summary:

This capstone project addresses critical operational challenges at Ganesh Medicals, a B2C retail pharmacy in Deoghar, Jharkhand, generating average monthly revenues of ₹16.53 lakh. The study analyzed primary data spanning 188 days of sales transactions (April 1 to October 5, 2025) comprising 65,343 records and 226 days of procurement data (April 1 to November 12, 2025), exported directly from the pharmacy's POS and inventory management systems.

Key Analytical Findings: ABC classification revealed extreme revenue concentration, with 22% of products (169 SKUs) contributing 80% of total revenue (₹79.4 lakh out of ₹99.2 lakh), while 78% of inventory items generated only 20% of sales. Return rate analysis identified systematic inefficiencies, with total returns of ₹1.78 lakh (2.75% of gross sales), including two products with 100% return rates and ten products with return rates exceeding 30%, indicating complete demand forecasting failures for specialized and surgical items.

Seasonal demand analysis documented pronounced volatility, with sales increasing 54.6% from July (13,600 units) to September peak (23,195 units). Despite strong overall procurement-sales correlation (0.86), timing gaps emerged during transition periods, with procurement lagging sales growth by 3.5 percentage points during the critical July-August surge. True demand reconstruction, accounting for documented stockouts, revealed that observed sales understated actual demand by 0.9% in April and 2.4% in July, confirming procurement stress during demand acceleration phases rather than at absolute peaks.

Supplier performance analysis uncovered substantial cost heterogeneity masked by uniform headline discount rates (4.75-5.0%). Effective rate calculations, incorporating both trade discounts and free-quantity schemes, revealed an 89% cost dispersion between the most expensive (R.P. Distributors: ₹100.78/unit) and least expensive (Kedia Drug Agency: ₹53.25/unit) suppliers. At current procurement volumes (118,312 units), this represents potential annual savings of ₹16.05 lakh through systematic cost-based supplier selection.

Strategic Recommendations: Implementation of differentiated inventory policies based on ABC classification with velocity-based reorder points for 169 high-value SKUs; seasonal anticipation procurement using 1.4x multipliers for monsoon-related medicines during August-September; and supplier selection based on composite scorecards emphasizing effective rates rather than promotional schemes.

Projected cumulative impact: ₹8.1-11.6 lakh in annual savings through reduced returns (₹0.8-0.9 lakh), captured seasonal revenue (₹1.5-2.0 lakh), and procurement optimization (₹5.8-8.7 lakh), representing 11-15% margin improvement and transitioning Ganesh Medicals from reactive, heuristic-based operations to predictive, analytically-supported inventory and procurement management.

Detailed Explanation of Analysis Process / Methods:

2.1 Data Collection Process

Primary data was collected through direct access to Ganesh Medical's billing software database, with Mr. Kedia (pharmacy owner) providing authorized access. The data collection spanned from April 1, 2025 to October 5, 2025 for sales transactions and extended to November 2025 for procurement records to capture forward planning

Data was exported as Excel files (.xlsx) directly from:

- POS system for sales transactions
- Inventory management database for procurement records

Data authenticity was verified through official documentation for inventory.

The pharmacy owner provided authorized access to the billing software database. The data was exported as excel files directly from the POS system and inventory database, ensuring automated data capture accuracy.

LINK:

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2.2 Data Cleaning and Preprocessing

2.2.1 Anomalies encountered and resolved

Data cleaning was performed systematically to ensure analytical reliability:

Daily Aggregate Rows: Dataset contained daily summary rows with totals that were removed, resulting in empty rows and null values requiring deletion. Reduction from 65,530 to 65,343 entries (187 aggregate rows removed).

Negative Quantities: Qty column had values ranging from -56 to 150. Negative values were segregated into returns (marked with S/Re) and stockouts (marked with shortfall), enabling true demand reconstruction.

Item Name Standardization: Inconsistent abbreviations (TAB vs TABLET, INJ vs INJECTION) and dosage formatting variations were standardized using text normalization to ensure accurate product matching between sales and procurement datasets.

Missing Batch Numbers: Some transactions lacked batch information, limiting batch-level turnover analysis. These entries were flagged but retained for product-level analysis.

Outlier Detection: Box plots were generated for Qty and Amount fields. Extreme values were verified against original records rather than automatically removed, as high-quantity purchases for surgical items were legitimate business transactions.

Tools used: Microsoft Excel for initial cleaning, Python Pandas for systematic text standardization and outlier flagging, with all transformations documented for reproducibility.

2.3 Analytical Methods Mapped to Problem Statements

2.3.1 Methods for PS1: Inventory Inefficiency and Expiry-Linked Losses

2.3.1.1 : ABC Classification with Sales Velocity

Definition: ABC classification is a value-based inventory categorization technique derived from the Pareto principle (80/20 rule), segmenting products by their contribution to total revenue. Products were rank-ordered by individual revenue contribution (unit price × quantity sold), and cumulative revenue percentages were calculated. Classification thresholds were applied as follows: A-category items (top 20% of SKUs contributing approximately 80% of cumulative revenue), B-category items (next 30% of SKUs contributing

approximately 15% of revenue), and C-category items (remaining 50% of SKUs contributing approximately 5% of revenue).

This revenue-based classification was augmented with a sales velocity metric to create a two-dimensional inventory prioritization matrix. Sales velocity was calculated for each SKU as:

Formula:

Sales Velocity = Total Units Sold / Study Days;

Cumulative Revenue % = (Cumulative Revenue / Total Revenue) × 100

Products were then classified into velocity bands: high-velocity (>1.0 units/day), medium-velocity (0.2-1.0 units/day), and low-velocity (<0.2 units/day). The intersection of ABC categories and velocity bands produced nine inventory segments (**A-High, A-Medium, A-Low, B-High, B-Medium, B-Low, C-High, C-Medium, C-Low**), each requiring differentiated management strategies.

Justification: Pure revenue-based ABC classification, while identifying value concentration, can produce misleading inventory policies if applied without velocity considerations. A high-value product with minimal turnover (A-category but low-velocity) poses expiry risk and capital lock-up, warranting stock reduction despite its revenue contribution per transaction. Conversely, high-velocity products even in lower value categories (C-category but high-velocity) require consistent availability to prevent stockout-driven customer dissatisfaction and brand switching.

This dual-metric approach addresses Problem Statement 1 (inventory inefficiency and expiry-linked losses) more comprehensively than single-dimension methods. It identifies:

- **High-priority items** (A-category, high-velocity): Require tight availability management, frequent reorder points, and safety stock to prevent revenue loss from stockouts.
- **Expiry-risk items** (A/B-category, low-velocity): High per-unit value but slow movement creates capital inefficiency and expiry vulnerability—candidates for stock reduction or demand-stimulation strategies.
- **Dead-stock candidates** (C-category, low-velocity): Minimal revenue contribution with minimal movement—prime targets for discontinuation or shift to on-demand ordering.

2.3.1.2 : Stockout Impact Analysis

Definition: Stockout impact analysis reconstructs unfulfilled demand by leveraging explicit

stockout documentation in the sales dataset. The POS system records instances where customer requests cannot be fulfilled due to inventory unavailability, marked with "shortfall" notation in the Party Name field and recorded as negative quantities representing units demanded but not sold.

For each SKU, the stockout rate was calculated as:

$$\text{Formula: Stockout Rate} = (|\text{Stockout Qty}| / (\text{Sales Qty} + |\text{Stockout Qty}|)) \times 100$$

True demand was reconstructed as:

$$\text{True Demand} = \text{Observed Sales} + |\text{Documented Stockouts}|$$

Justification: Conventional sales analysis treats observed sales as equivalent to customer demand, an assumption valid only when inventory availability is unconstrained. In resource-limited retail contexts like Ganesh Medicals, stockouts are recurring events that create a demand-visibility gap: when shelves are empty, zero sales are recorded, masking latent customer interest. This leads to systematic underestimation of true demand, particularly for products with volatile or seasonal patterns where procurement lags consumption.

The stockout reconstruction method addresses this by treating explicitly documented stockout events as demand signals. This approach is more rigorous than: (1) anecdotal stockout reports (which lack quantification), (2) lost-sales estimation models (which require assumptions about customer behavior and substitution patterns), or (3) backorder analysis (which this pharmacy doesn't systematically track). The method leverages a unique data asset—the pharmacy's practice of recording unfulfilled requests in the billing system for future procurement reference—to convert qualitative complaints ("we often run out of X") into quantitative demand metrics.

This directly addresses Problem Statement 1 in two ways. First, it identifies products with chronic availability issues where stockouts cause revenue leakage (high-value items with high stockout rates represent immediate profit recovery opportunities). Second, it reveals procurement-demand mismatches where low observed sales might reflect limited supply rather than limited demand, preventing erroneous conclusions that slow-selling items are unwanted (when they might be perpetually understocked).

Alternative demand reconstruction approaches were considered: regression-based demand forecasting requires multi-year historical data unavailable in this single-year study; customer survey methods introduce recall bias and sampling limitations; market basket analysis of

substitution patterns requires point-of-sale basket data not captured in the current system. The documented stockout approach uniquely provides product-specific, transaction-level demand data without additional data collection or modeling assumptions.

2.3.2 Methods for PS2: Seasonality-Driven Demand Misalignment

2.3.2.1 : Month-over-Month Growth Comparison

Definition: Month-over-Month (MoM) growth analysis quantifies the rate of change in sales and procurement volumes across consecutive months, enabling direct comparison of demand signals and procurement responses. For each month, aggregate sales quantities and purchase quantities were calculated, and growth rates were computed as:

Formula: MoM Growth = $((\text{Month}_N - \text{Month}_{N-1}) / \text{Month}_{N-1}) \times 100$;

Separate growth series were generated for sales and procurement, creating parallel time series spanning the overlapping observation period (April-October). Pearson correlation coefficient (r) was calculated between the two growth series to quantify overall alignment strength. Divergence greater than $\pm 10\%$ was observed during July–August transition

The analysis distinguishes between absolute volume alignment (comparing raw unit counts) and growth rate alignment (comparing percentage changes). While absolute volumes may differ systematically (procurement often exceeds sales to build inventory), growth rates should move directionally together if procurement responds to demand signals. Correlation of growth rates provides this directional responsiveness measure.

Justification: This method was chosen because growth rates provide a normalized and comparable measure across months with varying volumes. Unlike absolute values, percentage changes capture both magnitude and direction of demand shifts, making them suitable for identifying whether procurement responds to changing demand patterns. Given the availability of only a single-year dataset, this approach offers a practical and interpretable way to assess seasonality-driven alignment without requiring multi-year time series models. It directly supports the analysis of whether procurement decisions are reactive or aligned with observed demand trends.

2.3.2.2 : True Demand Reconstruction with Stockout Adjustment

Definition: True demand reconstruction, introduced in Section 2.3.1.2 for stockout analysis, is extended here to seasonal context by calculating month-by-month true demand figures incorporating documented stockouts:

$$\text{True Demand_Month} = \text{Observed Sales_Month} + |\text{Stockout Quantity_Month}|$$

The difference between true demand and observed sales represents the demand-visibility gap caused by inventory constraints. This gap percentage is calculated monthly as:

$$\text{Demand Gap \%} = [|\text{Stockout Quantity}| / \text{True Demand}] \times 100$$

Months with high demand gaps indicate periods where stockouts are suppressing observed sales, making procurement based on sales history systematically inadequate. Seasonal patterns in the demand gap reveal whether stockouts concentrate in specific calendar periods (suggesting predictable seasonal procurement failures) or are randomly distributed (suggesting systemic inventory inadequacy).

Justification: This method was chosen because observed sales alone do not fully capture demand during periods of stockouts, particularly in peak months. By incorporating stockout quantities, the analysis provides a more accurate representation of seasonal demand fluctuations. It helps identify whether procurement decisions are based on incomplete demand signals and highlights periods where inventory shortages lead to underestimation of demand. This is especially useful in a single-year dataset where traditional seasonal decomposition methods are less reliable.

2.3.3 Methods for PS3: Supplier Variability & Procurement Constraints

2.3.3.1 : Effective Rate Calculation

Definition: Effective rate calculation determines the true cost per unit procured after incorporating all cost-reduction mechanisms: cash discounts, trade discounts, and free-quantity promotional schemes. The formula accounts for both price reductions and quantity augmentation:

$$\text{Formula: Effective Rate} = (\text{Rate} \times (1 - \text{Discount\%}) \times \text{Qty}) / (\text{Qty} + \text{Free Qty})$$

Where:

- Rate = invoice unit price before discount
- Discount% = trade discount percentage offered
- Qty = units paid for
- Free Qty = promotional units received at no charge

This method differs from simple discounted price by incorporating the impact of free quantities, which significantly reduce the actual cost per unit.

Effective rates were calculated for all procurement transactions and aggregated at the supplier level using the median to reduce the influence of outliers from high-value specialty items.

Justification: This method was chosen because supplier comparison based only on discount percentages can be misleading. Free-quantity schemes often contribute more to cost reduction than nominal discounts. By incorporating both components, effective rate provides a more accurate and comparable measure of procurement efficiency across suppliers. It enables identification of cost-effective suppliers and supports better procurement decision-making.

2.3.3.2 : Supplier Performance Scorecard with multicriteria weighting

Definition: The supplier performance scorecard synthesizes multiple evaluation dimensions into a single composite score enabling objective vendor ranking and selection. Three primary criteria were selected based on business impact assessment.

- **Discount Percentage (30% weight):** Direct cash cost reduction
- **Free Quantity Percentage (20% weight):** Indirect cost reduction through quantity bonuses
- **Price Competitiveness (30% weight):** Market positioning relative to peer suppliers

Each criterion was normalized to a 0–100 scale using min-max normalization to ensure comparability across different units. For discount and free quantity, higher values resulted in higher scores, while for price competitiveness; lower effective rates were assigned higher scores.

The composite score for each supplier was calculated as:

Composite Score = (Discount Score × 0.30) + (Free Quantity Score × 0.20) + (Price Competitiveness Score × 0.30)

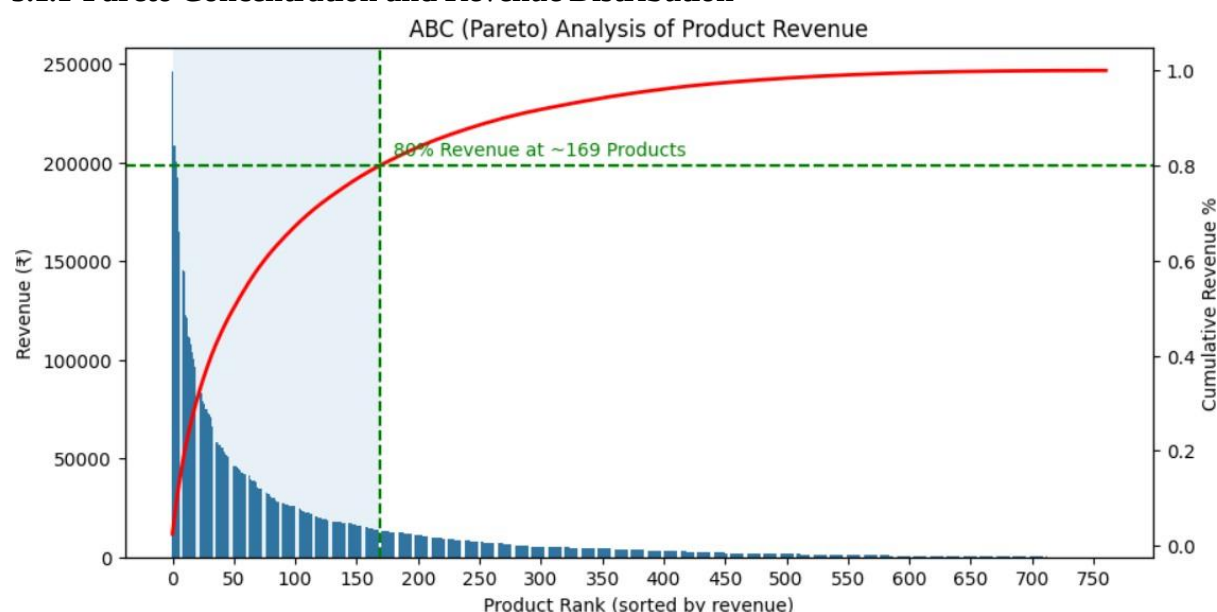
Suppliers were then ranked based on their composite scores to identify high-performing and low-performing vendors.

Justification: This method was used because evaluating suppliers based on a single metric, such as discount or price alone, can be misleading. The scorecard provides a balanced and structured evaluation by incorporating multiple dimensions of supplier performance. It enables objective comparison across suppliers and supports more informed procurement decisions by identifying vendors that offer the best overall value rather than relying on isolated metrics.

Results and Findings

3.1 Findings for Problem Statement 1: Inventory Inefficiency and expiry linked losses

3.1.1 Pareto Concentration and Revenue Distribution



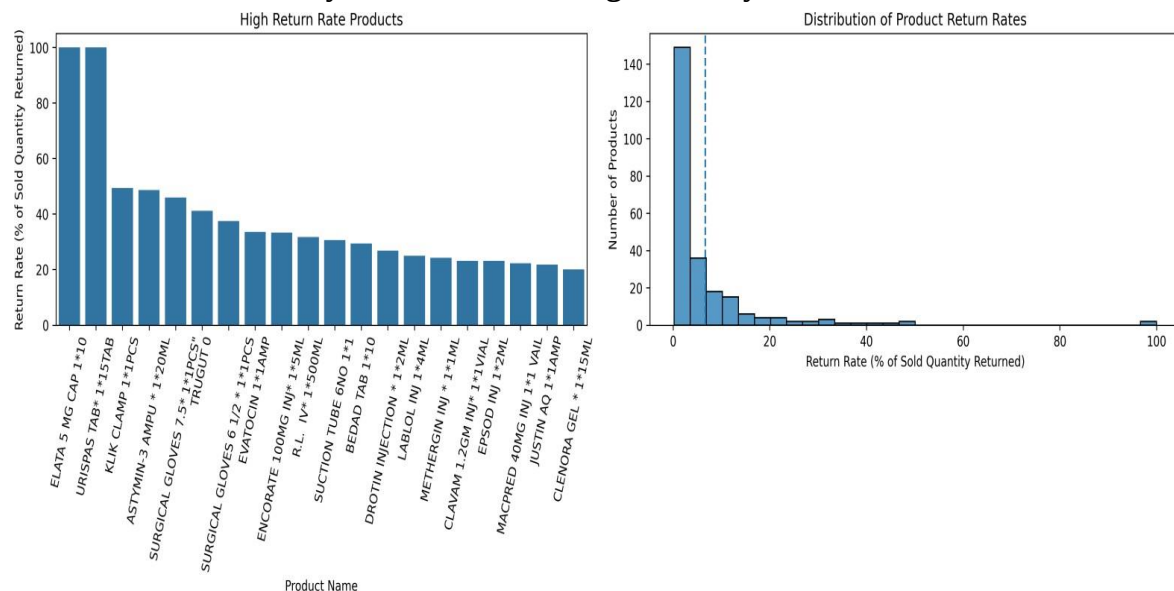
Graph 3.1.1: Pareto Concentration with Stockout Overlay

Key Findings:

- Revenue is highly concentrated: 169 products (22% of 766 total SKUs) generate Rs.79.4 lakh, representing 80% of total Rs.99.2 lakh revenue.
- An even narrower core dominates: fewer than 1% of products (approximately top 8 SKUs) contribute 15% of revenue (Rs.14.9 lakh), and just 0.26% (2 SKUs) account for 5% (Rs.5.0 lakh)
- The majority (597 products, 78% of SKUs) contribute only Rs.19.8 lakh (20% of revenue), creating a steep long-tail distribution

Business Interpretation: This distribution confirms a strong Pareto (80/20) pattern in the product portfolio. The results highlight that applying uniform inventory management across all 766 SKUs leads to inefficient resource allocation. High-impact A-category products require tighter availability control to prevent revenue loss from stockouts, while the large number of low-impact C-category products occupy shelf space and tie up working capital with minimal financial contribution. These low-value items are therefore key candidates for stock rationalization and return-risk reduction.

3.1.2 Return Rate Analysis and Slow Moving inventory



Graph 3.1.2: Top 20 Products by Return Rate(% of Sold qty which was returned:High Risk and Overall Distribution

Quantified Results:

Product	Return Rate	Value Impact
ELATA 5 MG CAP	100%	Total procurement returned
URISPAS TAB	100%	Total procurement returned
KLIK CLAMP	49%	Nearly half returned
SURGICAL GLOVES 7.5	30-45%	Systematic over- ordering

TRUGUT 0	41%	High return — significant loss
SURGICAL GLOVES 6	38%	High return — significant loss
EVATOCIN 1*1	34%	High return — significant loss
ENCORATE 100MG INJ* 1*5ML	33%	High return— significant loss
R.L. IV* 1*500ML	32%	High return — significant loss
SUCTION TUBE 6NO 1*1	31%	High return — significant loss
BEDAD TAB 1*10	30%	Moderate-high return
DROTIN INJECTION * 1*2ML	27%	Moderate-high return
LABLOL INJ 1*4ML	26%	Moderate-high return
METHERGIN INJ * 1*1ML	25%	Moderate-high return
CLAVAM 1.2GM INJ* 1*1VIAL	24%	Moderate-high return
EPSOD INJ 1*2ML	24%	Moderate-high return
MACPRED 40MG INJ 1*1 VAIL	23%	Moderate return
JUSTIN AQ 1*1AMP	22%	Moderate return
CLENORA GEL * 1*15ML	20%	Moderate return

Key Findings:

The return rate analysis reveals that return activity is highly concentrated in a small subset of products. Among 766 total SKUs, the top 20 products exhibit return rates ranging from 20% to 100%, with two products—ELATA 5 MG CAP and URISPAS TAB—showing complete return of procured inventory (100% return rate), indicating total demand failure.

The majority of high-return products fall within the 20–50% return range, suggesting systematic over-procurement rather than isolated anomalies. These include products such as KLIK CLAMP (49%), TRUGUT (41%), SURGICAL GLOVES (various sizes: 30–45%), and multiple injectable medications with return rates between 30–35%.

At an aggregate level, total returns across all products amount to 2,914 units valued at approximately ₹1.78 lakh, representing an overall return rate of 2.75% of total sales quantity.

However, this aggregate figure masks significant concentration: only 127 products experienced returns, while 639 products (83.4% of SKUs) had zero returns. As highlighted in the top 20 product analysis and further supported by the distribution plot (Figure 3.1.2-rhs), return activity is highly concentrated in a small subset of products, while the majority of items exhibit negligible return rates.

Distribution Insights:

The distribution of return rates across all products (Figure 3.1.2, right panel) shows a highly right-skewed pattern. Most products exhibit very low or negligible return rates, clustered near 0–5%, while a small number of products display significantly higher return rates extending up to 100%.

This skewed distribution indicates that inventory inefficiency is driven by a limited set of problematic products rather than systemic failure across the entire inventory. The presence of extreme outliers (100% return products) further highlights critical gaps in demand estimation for specific SKUs.

Observed Patterns in High-Return Products

Analysis of the top 20 high-return products reveals consistent category-level patterns:

- **Surgical and medical supplies** (e.g., gloves, clamps, suction tubes) show return rates between 30–49%, indicating procurement based on bulk ordering or anticipated demand that did not materialize.
- **Specialty injectables and hospital-use items** exhibit return rates in the 25–35% range, suggesting stocking decisions driven by irregular or uncertain patient demand.
- **Therapeutic medications with 100% return rates** indicate complete mismatch between procurement and actual prescription patterns.

Velocity–Return Relationship

Cross-analysis with sales velocity indicates that high-return products are predominantly low-velocity items. Products with slower movement are more likely to remain unsold for extended periods, increasing the likelihood of returns or expiry.

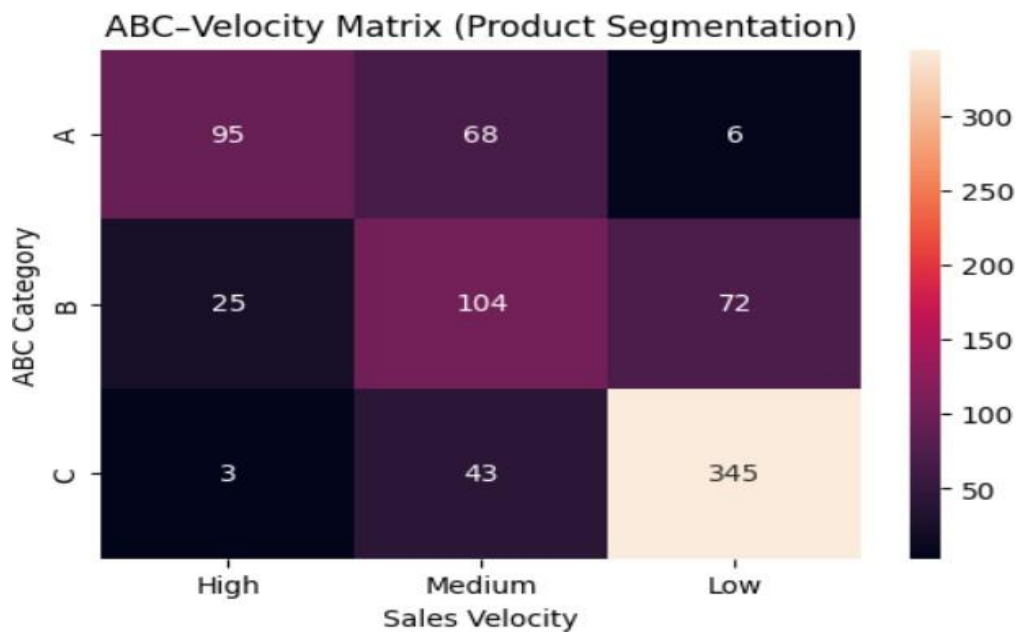
This establishes a strong inverse relationship between sales velocity and return rate, confirming that slow-moving inventory is the primary driver of return-related inefficiencies.

Business Interpretation: The results strongly support Problem Statement 1 regarding inventory inefficiency and expiry-linked losses. The concentration of returns in a small subset of low-velocity products indicates that current procurement practices do not adequately differentiate between high-demand and low-demand items.

While the majority of products operate efficiently with negligible returns, a limited number of poorly aligned procurement decisions result in disproportionately high return rates. The presence of products with complete return (100%) represents instances where procurement decisions were entirely disconnected from actual demand.

Overall, the findings highlight that inventory inefficiency is not a system-wide issue but a targeted problem affecting specific categories of products, particularly those with low demand visibility and irregular consumption patterns.

3.1.3 ABC–Velocity Matrix Analysis for Inventory Segmentation



Graph 3.1.3: ABC–Velocity Matrix Analysis for Inventory Segmentation

Key Findings:

- A-category products are primarily concentrated in high and medium velocity segments (95 high, 68 medium), indicating that most high-revenue products are also fast-moving and contribute consistently to sales.
- Only a small number of A-category products (6 SKUs) fall into the low-velocity segment, highlighting a limited but critical set of high-value items at risk of slow movement or expiry.
- B-category products show a distributed pattern across all velocity segments, with the highest concentration in medium (104) and low velocity (72), suggesting moderate demand variability and less predictable consumption patterns.
- C-category products are overwhelmingly concentrated in the low-velocity segment (345 SKUs), confirming that the majority of low-value products are also slow-moving.
- High-velocity C-category products are negligible (3 SKUs), indicating that low-value items rarely contribute significantly to frequent demand.

Business Interpretation: The ABC–velocity matrix highlights that inventory performance varies significantly across product segments and should not be managed uniformly.

High-value A-category products are largely fast-moving, making them critical for revenue generation and requiring strict availability management to avoid stockouts. However, the presence of a small number of low-velocity A-category products indicates potential expiry risk even within high-value items, necessitating closer monitoring and controlled stocking.

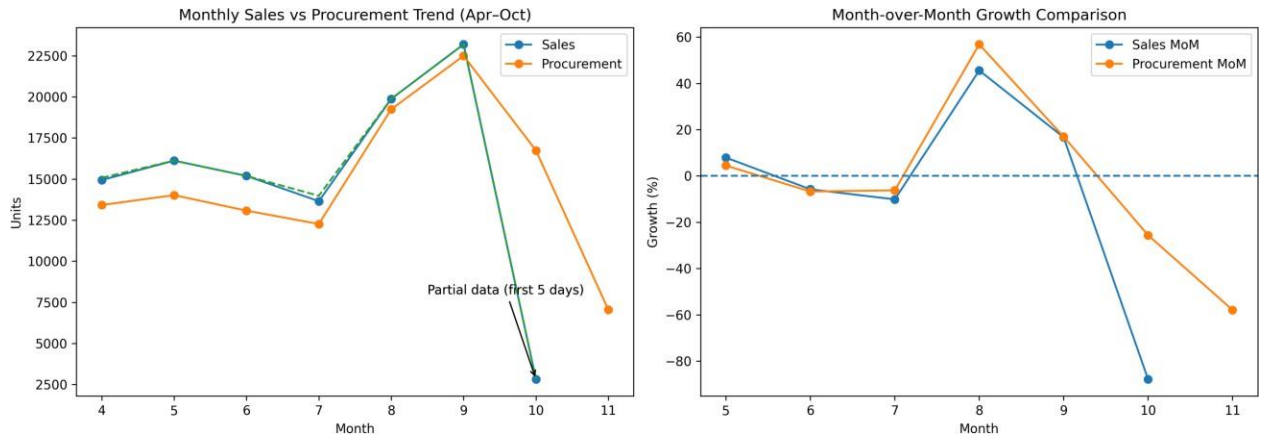
The dominance of low-velocity C-category products confirms that inventory inefficiency is primarily driven by the long tail of low-value items. These products contribute minimally to revenue while occupying storage space and tying up working capital, making them key targets for stock rationalization and reduced procurement frequency.

B-category products, with their spread across velocity segments, require balanced inventory policies, as their demand patterns are less stable and more sensitive to fluctuations.

Overall, the analysis reinforces that differentiated inventory strategies based on both value and movement are essential to improve efficiency and reduce wastage.

3.2 Findings for Problem Statement 2: Seasonality Driven Demand Misalignment

3.2.1 Seasonal Demand Patterns and Procurement Response Analysis



Graph 3.2.1: Monthly Sales and Procurement Trends with Month-over-Month Growth Comparison

Quantified Seasonal Pattern:

Month	Units Sold	MoM Change	Pattern
April	15,000	-	Baseline
May-June	~15,500	+3.3%	Stable
July	13,600	-12.3%	Dip
August	19,800	+45.6%	Surge
September	23,195	+17.1%	Peak

Key Findings:

- The combined visualization shows a **stable demand pattern during April–June** (approximately 15,000–15,600 units), indicating a consistent baseline level of sales before seasonal variation begins.
- A **decline in July (13,600 units)** marks a pre-season dip, followed by a **sharp surge in August (+45.6% MoM)** and continued growth into September (23,195 units), establishing a clear seasonal demand pattern.
- The **highest rate of change occurs in August**, not September, indicating that demand acceleration is most intense during the transition into the peak season rather than at the peak itself.

- Procurement trends broadly follow the direction of sales, as seen in the left panel, and this is supported by a **strong positive correlation ($r = 0.86$)** between sales and procurement MoM growth rates.
- However, **misalignment is evident during transition periods:**
 - During the **July–August surge**, procurement growth (+42.1%) lags behind sales growth (+45.6%), indicating delayed response during demand acceleration.
 - During the **post-peak phase**, procurement does not decline as sharply as sales, suggesting over-adjustment and potential excess inventory buildup.
- The inclusion of **true demand (dashed line)** shows that actual demand slightly exceeds observed sales during certain months (e.g., July), indicating that stockouts partially suppress visible demand.
- The **sharp drop in October** is not indicative of actual demand collapse but reflects **partial data (first 5 days only)** and is excluded from seasonal pattern interpretation.

Business Interpretations:

The merged analysis of absolute trends and growth rates reveals that while procurement decisions are directionally aligned with demand, they are not sufficiently responsive to rapid changes in demand patterns.

The most critical inefficiencies occur during **transition periods**, particularly during demand acceleration. The sharp increase in August represents a demand shock, where procurement lags behind sales growth, increasing the risk of stockouts and lost sales opportunities. This indicates reliance on reactive procurement strategies rather than anticipatory planning.

Similarly, during the **post-peak phase**, procurement adjustments are slower than demand decline, leading to excess inventory accumulation. This suggests that procurement decisions are not dynamically calibrated to demand normalization, increasing the risk of overstocking and potential expiry losses.

The true demand analysis further reinforces that observed sales alone underestimate actual demand during key transition periods, particularly when stockouts occur. This implies that procurement decisions based solely on historical sales data may systematically underpredict demand during critical phases.

Overall, the findings confirm that seasonality-driven demand fluctuations are not fully integrated into procurement planning. While the system performs adequately under stable conditions, it fails to adapt efficiently during periods of rapid change, leading to both understocking during surges and overstocking during declines.

3.3 Findings for Problem Statement 3: Supplier Variability

3.3.1 Supplier Performance Heterogeneity



Graph 5: Top 5 Supplier Comparison

Quantified Supplier Comparison:

Supplier	Discount score	Free score	Qty	Price score	Composite Score
KEDIA DRUG AGENCY	95.0	27.2		100.0	63.94
B.K. ENTERPRISES	100.0	60.1		90.6	69.20
ARYA DISTRIBUTORS	100.0	21.9		71.4	55.81
HINDUSTAN PHARMA TRADERS	100.0	28.7		28.6	44.36
R.P. DISTRIBUTORS	100.0	38.4		0.0	37.68

Key Findings: The supplier performance scorecard reveals significant variation in overall supplier value despite nearly uniform discount scores across all suppliers.

B.K. Enterprises emerges as the top-performing supplier with the highest composite score (69.20), driven by strong free quantity offerings (60.08%) and high price competitiveness. Kedia Drug Agency follows with a composite score of 63.94, supported by the highest price competitiveness score (100), despite relatively lower free quantity benefits.

Arya Distributors demonstrates moderate performance (55.81), benefiting from relatively strong pricing but lower promotional support. Hindustan Pharma Traders shows lower overall performance (44.36) due to weaker price competitiveness despite reasonable discounts.

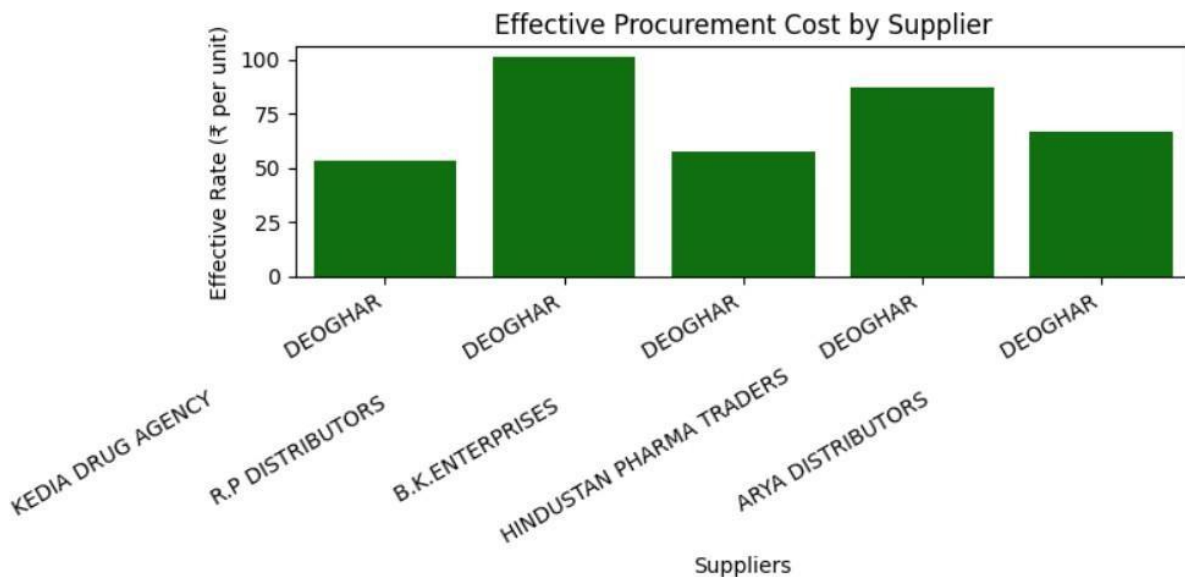
R.P. Distributors ranks as the least efficient supplier (37.68), primarily due to the lowest price competitiveness score (0), indicating significantly higher procurement costs compared to other suppliers.

Business Interpretation: The analysis highlights that supplier performance cannot be evaluated solely on discount percentages, as all suppliers offer nearly identical discount levels. Instead, significant differences arise from variations in free quantity schemes and effective pricing.

B.K. Enterprises and Kedia Drug Agency provide the highest overall value through a combination of competitive pricing and promotional benefits, making them more suitable for procurement prioritization. In contrast, suppliers such as R.P. Distributors exhibit poor cost efficiency despite offering similar discounts, indicating potential inefficiencies in current procurement decisions.

The wide variation in composite scores demonstrates substantial opportunity for cost optimization by shifting procurement toward high-performing suppliers. Implementing a structured, multi-criteria supplier evaluation framework can significantly improve procurement efficiency and reduce overall costs.

3.3.2 Effective Rate Analysis



Graph 6: Effective Rate Comparison (Top 5 Suppliers)

Key Findings:

The effective rate analysis reveals substantial variation in procurement costs across suppliers. Kedia Drug Agency offers the lowest median effective rate (₹53.25 per unit), making it the most cost-efficient supplier. In contrast, R.P. Distributors has the highest effective rate (₹100.78 per unit), representing an 89% higher cost compared to Kedia.

Mid-tier suppliers such as B.K. Enterprises and Arya Distributors fall between these extremes, indicating moderate cost competitiveness.

Business Interpretation: The wide variation in effective rates highlights significant inefficiencies in supplier selection. While some suppliers offer competitive pricing, others incur substantially higher procurement costs despite similar discount structures.

This indicates that procurement decisions may not be fully aligned with cost efficiency and could be influenced by non-price factors such as supplier relationships or ordering convenience.

The analysis demonstrates the importance of evaluating suppliers based on effective rate rather than discount percentage alone. While Kedia Drug Agency is the most cost-efficient supplier, the scorecard analysis shows that overall supplier performance also depends on promotional benefits and pricing consistency.

Interpretation of Results and Recommendations

4.1 Interpretation: Synthesis of Cross-Problem Insights

The three problem statements are closely interconnected through working capital dynamics and decision-making inefficiencies. The analysis shows that **169 SKUs (22%) contribute 80% of total revenue**, while **597 SKUs (78%) contribute only 20%**, indicating a highly skewed product portfolio. This imbalance leads to inefficient capital allocation, where excessive investment in low-impact items reduces the availability of funds for high-demand products.

Inventory inefficiency (PS1) is reflected in **₹1.78 lakh worth of returns**, primarily concentrated in slow-moving and specialized items. At the same time, seasonality misalignment (PS2) creates **timing gaps during peak demand periods (Aug–Sep)**, where true demand exceeds observed sales by up to **2.4%**, resulting in missed revenue opportunities. Supplier variability (PS3) further compounds these issues, with significant differences in effective procurement cost, leading to inefficient purchasing decisions and additional working capital strain.

The underlying root cause across all three problems is the reliance on **reactive, experience-based decision-making** rather than a structured, data-driven approach. Although a strong **correlation of 0.86 between sales and procurement trends** indicates that the system responds to demand signals, the presence of **100% return rates and procurement timing gaps** highlights the lack of predictive planning. This confirms the need for an integrated, analytics-driven framework for inventory and procurement decisions.

4.2 Actionable Recommendations

4.2.1 Problem Statement 1: Inventory Inefficiency and Expiry-Linked Losses

The analysis reveals a highly concentrated revenue structure where **169 SKUs (22%) contribute nearly 80% of total revenue (₹79.4 lakh out of ₹99.2 lakh)**, while the remaining 78% of products contribute only 20%, indicating inefficient allocation of working capital. Additionally, return analysis shows **2.75% of total units (₹1.78 lakh)** being returned, with certain products exhibiting **100% return rates**, reflecting a mismatch between

procurement and actual demand. The high **coefficient of variation (105%)** further indicates demand volatility, making rule-based or experience-driven purchasing ineffective. These findings confirm that the pharmacy lacks differentiated inventory control, leading to overstocking of low-value items and stockouts of critical medicines.

Recommendations:

- Maintain current digital record-keeping in POS and inventory systems, which enabled this analysis
- Continue documenting stockouts and returns with clear notation (shortfall, S/Re), as this data proved invaluable for true demand reconstruction
- **Implement ABC-based inventory classification:**
Categorize products into A (high value, high demand), B (moderate), and C (low value) categories based on revenue contribution and sales velocity. Focus strict monitoring and frequent replenishment on A-items, while limiting stock for C-items.
- **Introduce reorder point system using sales velocity:**
Calculate reorder levels based on average daily sales and lead time to ensure timely replenishment of fast-moving medicines and prevent stockouts.
- **Reduce procurement of slow-moving and high-return items:**
Identify products with high return rates or low sales velocity and either minimize their stock levels or shift to on-demand ordering.
- **Align procurement with demand variability:**
Adjust purchase quantities based on observed demand fluctuations instead of fixed or bulk ordering practices.
- **Avoid uniform stocking policies across all SKUs:**
Treat all products differently based on their contribution and movement instead of applying the same stocking logic to all items.

4.2.2 Problem Statement 2: Seasonality-Driven Demand Misalignment

The analysis highlights strong seasonal demand variation, with a **54.6% increase in sales from July (13,600 units) to September (23,195 units)**. Although procurement shows a relatively strong **correlation of 0.86 with sales**, indicating general alignment, there is a **lag during peak growth periods**, leading to stock shortages when demand surges. Additionally, true demand analysis (including stockouts) indicates unmet demand during transition phases,

confirming that procurement decisions are reactive rather than predictive. This results in both lost sales during peak demand and excess inventory during off-peak periods.

Recommendations:

- **Adopt seasonal demand forecasting techniques:**
Use moving averages or trend-based methods to identify seasonal patterns and predict demand for upcoming months.
- **Advance procurement before peak seasons:**
Increase stock levels of high-demand medicines 4-6 weeks before expected seasonal peaks (e.g., monsoon or flu season).
- **Create seasonal procurement multipliers by product category:** For monsoon-related medicines (anti-malarials, antibiotics), set procurement at 1.4x normal levels for August delivery. For general medicines, 1.2x multiplier. This addresses the +45.6% Aug surge more proactively.
- **Use true demand instead of observed sales:**
Incorporate stockout quantities along with actual sales to estimate real demand and improve forecasting accuracy.
- **Develop dynamic procurement schedules:**
Adjust ordering frequency and quantity based on seasonal trends instead of following fixed monthly cycles.
- **Avoid reactive purchasing behavior:**
Shift from short-term, experience-based ordering to data-driven, forward-looking procurement planning based on demand trends.

4.2.3 Problem Statement 3: Supplier Variability and Procurement Constraints

The supplier analysis indicates that **three suppliers contribute approximately 48% of total procurement value**, highlighting a high dependency on a limited supplier base. Despite similar discount rates (~4–5%), there is significant variation in actual procurement cost due to differences in free quantity schemes and pricing structures. This suggests that procurement decisions are currently influenced by visible discounts rather than true cost efficiency, leading to suboptimal supplier selection and inconsistent purchasing outcomes.

Recommendations:

- **Evaluate suppliers using effective cost (not just discount %):**
Calculate effective rate by incorporating both discounts and free quantities to identify the true cost advantage of each supplier.
- **Develop a supplier performance scorecard:**
Rank suppliers based on multiple criteria such as effective cost, reliability, delivery consistency, and trade benefits to support objective decision-making.
- **Diversify supplier base:**
Reduce dependency on a few suppliers by engaging with alternative vendors to improve negotiation power and reduce supply risk.
- **Standardize supplier selection criteria:**
Establish clear guidelines for choosing suppliers based on cost efficiency and service quality rather than informal judgment.
- **Avoid procurement decisions based solely on discounts:**
Do not prioritize suppliers offering higher discounts without considering overall value, as this may lead to higher effective costs.

END

OF

REPORT
